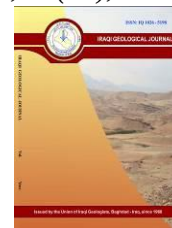




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Genetic Classification of Crude Oils from Cretaceous Reservoirs in the Ratawi Oil Field, South of Iraq: Molecular Evidence for Multiple Source Facies

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Abstract

The current study evaluated the genetic characterization and differentiation of oil families in the units of the Yamama, Nahr Umr, and Mishrif reservoirs of the Ratawi oil field, south of Iraq. The combination of bulk properties, stable carbon isotopes, and molecular biomarkers reveals a distinct pattern of hydrocarbon generation, migration, and accumulation, influenced by salt-induced tectonics in the study area. Based on Stable carbon isotopic and molecular fingerprints, two different oil sub-families were determined: (1) Sub-family I (Lower Cretaceous-Yamama A oil), which is mainly trapped in the unit of the Yamama reservoir, is interpreted to have originated from the Sargelu/Najmah generative pods. The carbon isotope value of ($\delta^{13}\text{C}$ approx -26.4‰) suggests a Cretaceous source origin, but all other molecular biomarkers refer to an Upper Jurassic source. This is indicated by a low $\text{C}_{28}/\text{C}_{29}$ sterane ratio (about 0.55) and a high amount of C_{27} steranes. So, this isotopic shift is interpreted as a high thermal maturity level, evidenced by a high Triaromatic Steroid ratio and high $\text{Ts}/(\text{Ts}+\text{Tm})$ ratios, causing carbon isotopic fractionation. This enriched the oil in heavy carbon isotopes ($\delta^{13}\text{C}$), masking the original Jurassic isotopic properties. Family II (Lower Cretaceous: Mishrif, Nahr Umr, Yamama B oils) has a constant Lower Cretaceous affinity. The biomarker ratios show higher $\text{C}_{28}/\text{C}_{29}$ sterane ratios (>0.61) and the presence of Oleanane [strongly anoxic (euxinic) depositional conditions], especially in the Nahr Umr crude oil samples. All oil samples were exposed to a highly reducing, anoxic depositional environment, as shown by low Pristane/Phytane ($\text{Pr}/\text{Ph} < 1.0$) and high Vanadium/Nickel ($\text{V}/\text{V}+\text{Ni}$) ratios, suggesting a marine carbonate source rock. Family, reflects a distal, inactive basinal environment. Whereas Family II, exhibits a transition toward a near-shore marine environment with increased terrigenous input, as indicated by a higher C_{29} sterane ratio (up to 42%) and higher Hopane/Sterane ratios. Several indicators, such as isomerization of C_{32} homohopanes, Triaromatic Steroid, $\text{C}_{27}\text{Ts}/\text{Tm}$, and CPI values close to 1.0, suggest that these oil sources have entered the main to late oil window.

Keywords: Carbon Isotopic; Biomarkers; Yamama Formation; Steranes; Thermal Maturity Level; Ratawi Oil Field; Iraq

1. Introduction

Integrated geochemical analyses, especially molecular biomarkers and stable carbon isotopes, offer a better understanding of the petroleum systems in the Ratawi oil field of the Zubair Sub-Zone, which is also known as the Gotnia Basin. Biomarkers give important data on the depositional environment, thermal maturity levels, and the age of potential source rocks. These help identify the origin of
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hydrocarbons, which is from the Middle Jurassic or Lower Cretaceous, and differentiate them (Pitman et al. 2004). Stable carbon isotope ^{13}C can identify the depositional environment and organic input, as well as potential fluid mixing during hydrocarbon migration through fault networks. Such geochemical studies are essential for reducing risk in exploration and development by enhancing predictions of reservoir charging mechanisms and fluid properties, especially with migration pathways. The combination of biomarker analysis and isotopic analysis is important for accurately defining petroleum systems in the Ratawi reservoirs (Pitman et al., 2004; Peters et al., 2005) (Figs. 1 and 2).



Fig. 1. Oil and gas map of Iraq, showing location of the studied oil field (Al-Khafaji et al., 2021)

The Ratawi Oil Field is one of the important hydrocarbon accumulations, located in the Zubair Sub-Zone (Gotnia Basin) of the Mesopotamian Foredeep (Aqrabi et al., 2010). The Ratawi structure is an anticline affected by deep-buried Infracambrian salt halokinesis, which has generated unique vertical pathways for fluids migrating from deep, thermally mature Middle Jurassic and Lower Cretaceous sources into the overlying reservoirs (Edgell, 1996; Aqrabi et al., 2010). The movement of the Infracambrian Hormuz Salt is the main influence on the petroleum systems in the Southern Mesopotamian Basin, especially the Trap Formation. The Ratawi anticline is the result of a deep-buried salt pillow or diapir pushing up. As salt rises, it fractures and faults the rocks above it (Aqrabi et al.,

2010). These faults act as pathways for the Middle Jurassic-Sargelu-sourced oils to migrate vertically into Cretaceous reservoirs (e.g., Yamama, Nahr Umr, and Mishrif formations).

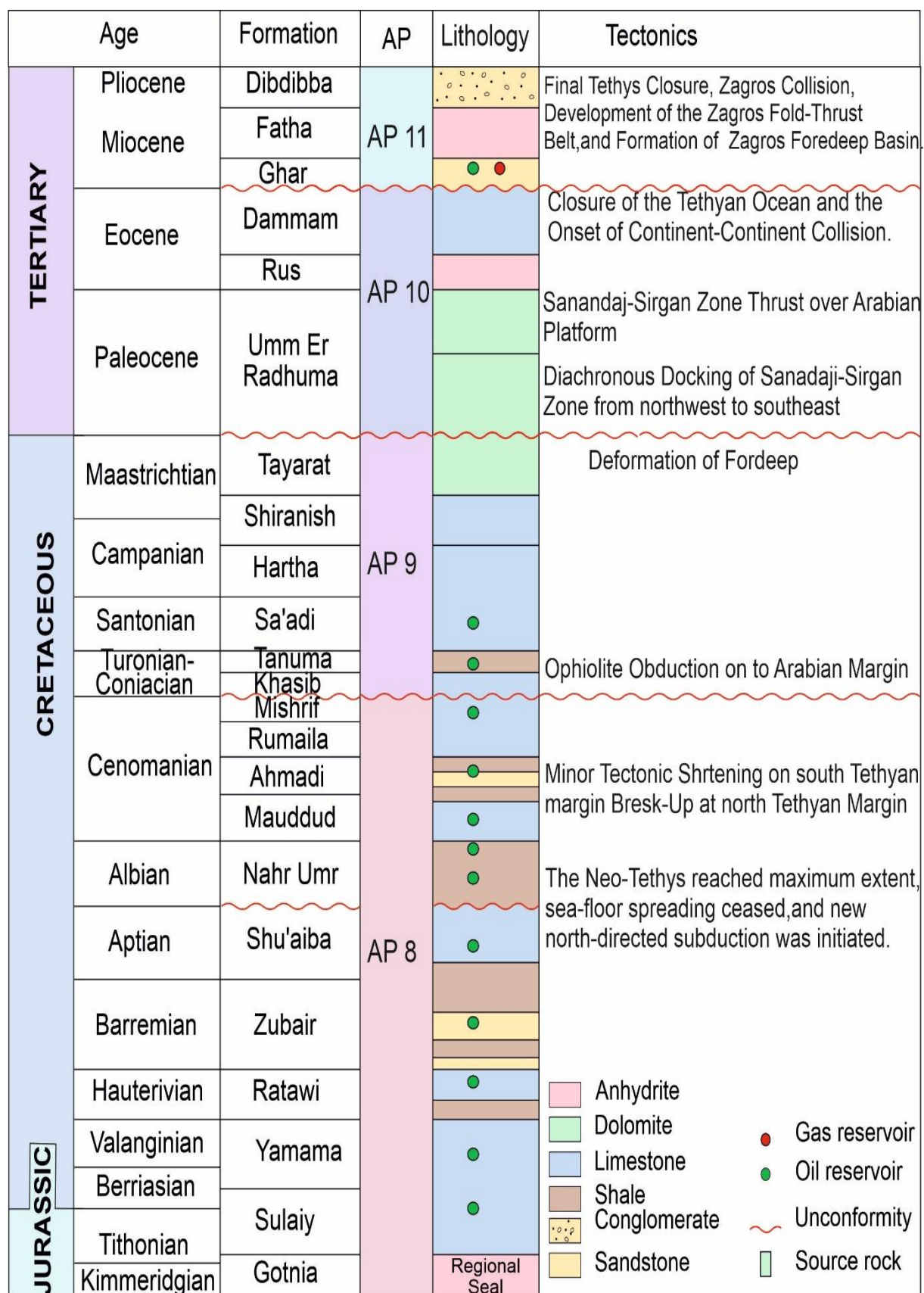


Fig. 2. General stratigraphic column of southern Iraq (Aqrawi et al., 2010)

The significance of these studies lies in their ability to establish genetic correlations between oils and their potential source rocks, thereby defining the efficiency and boundaries of a petroleum system (Pitman et al., 2004; Peters et al., 2005). The petroleum systems of southern Iraq have been the subject of much research, most notably the foundational work of Pitman et al. (2004). Al-Ameri (2015) studied the petroleum system and identified oil generation and migration in the south of the Mesopotamian Basin. They state that Middle Jurassic Sargelu source rocks are the main oil source in Iraq, with petroleum generation starting in the Late Cretaceous Period and peaking in the Late Miocene and Pliocene. In contrast, Abeed et al. (2013) and Al-Khafaji et al. (2021) have confirmed that the Sulaiy Formation is an excellent source rock and yields abundant oil in south Iraq.

The current study objectives include genetic classification of crude oil families in the Yamama, Nahr Umr, and Mishrif reservoirs of the Ratawi Oil Field, identification of the source rock facies, assessment of thermal maturity, and modelling of migration pathways influenced by the tectonic framework of the Zubair Sub-Zone. The conclusions of the current study also reduce exploration risk and enhance prospect evaluation.

2. Materials and Methods

In this study, the organic geochemical analysis was performed on 8 crude oil samples collected from various stratigraphic units: Yamama Formation (4 samples), Nahr Umr Formation (1 sample), and Mishrif Formation (3 samples). The analysis was developed to determine the genetic origin, thermal maturity, and depositional environment of fluids in the study area. All samples underwent organic geochemical analysis at GeoMark Research Company in Texas, USA.

Bulk Properties and Trace Element Analysis, such as Sulfur, Nickel (Ni), and Vanadium (V) contents, were calculated to identify the lithology and depositional environment of potential source rocks (for example, marine carbonates versus shale). The Vanadium/Nickel (V/Ni) ratio is a reliable indicator of the redox conditions during deposition. API Gravity is measured to assess the physical quality and thermal maturity levels of the oils across the Mesopotamian Basin. Stable carbon isotope ($\delta^{13}\text{C}$) analysis of the saturated and aromatic fractions was conducted to study oil-oil and oil-source rocks correlations to identify the depositional environment of the source rocks. Gas Chromatography (GC) analysis was used to identify the biomarker fingerprint of the n-alkane and isoprenoid distributions (such as Pristane/Phytane ratios). Also, to evaluate the degree of biodegradation, thermal maturity level, and oxic/anoxic condition of the paleoenvironment. Gas Chromatography-Mass Spectrometry (GC-MS) is an advanced technique used to identify specific biomarker parameters (Steranes and Terpanes). GC-MS is essential for identifying the type of organic matter (kerogen). Molecular ratios are used to estimate accurate thermal maturity levels and to identify migration pathways across the tectonic boundary by analysing minor molecular differences between samples.

3. Results and Discussions

3.1. Crude Oil Composition and Geochemical Indicators

The detailed molecular composition of crude oil is a key indicator of its origin, maturity, and biodegradation history (Tissot and Welte, 1984). The Sat/Aro ratio is used to determine biodegradation because a decrease in this ratio indicates microbial intake of saturates, which reduces oil quality (Tissot and Welte, 1984; Peters et al., 2005). Therefore, the Yamama oil sub-family (A) has a lower Sat/Aro ratio (1.03) than the sub-family (B), which has 1.18, while the Nahr Umr and Mishrif oil samples have 0.67 and 0.53, respectively, which are less than that of the Yamama oils. These show that the Yamama oils were of a younger source rock age and a higher maturation level, and more saturated than the oils of Nahr Umr and Mishrif, which were aromatic oils and originated from the Jurassic to Cretaceous

sequence. The Yamama sub-family (A) oils contain the lowest percentage of asphalt, compared to the oils of Nahr Umr and Mishrif, at 0.6%, while Yamama sub-family (B) oils have a relatively high percentage at 7.6%. Meanwhile, Nahr Umr and Mishrif oils contain even higher percentages of asphalt, at 7.2% and 12.5%, respectively, as shown in Fig. 3.

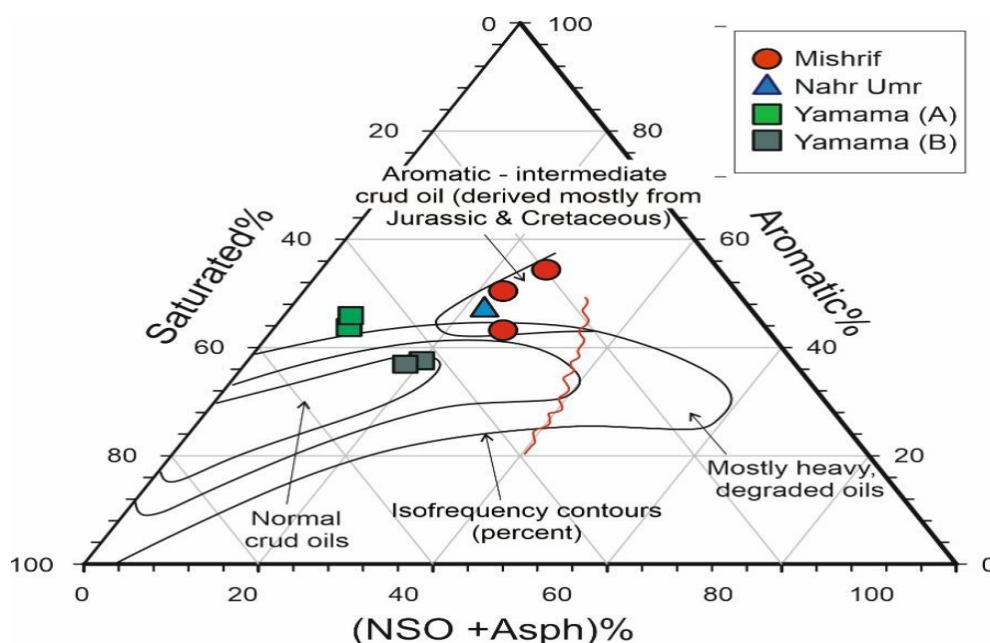


Fig. 3. A ternary diagram illustrates the gross composition of the studied crude oil samples for the construction of frequency contours; a triangle unit of 10: 10: 10 was used (Tissot and Welte, 1984)

3.2. Bulk Properties and Non-Biomarker Analysis

The API gravity is the most important physical indicator of crude oil properties and thermal maturity. A high API gravity typically indicates a higher thermal maturity or a higher gas-to-oil ratio (GOR), while lower API values may indicate a different genetic origin or the influence of secondary processes such as water washing or biodegradation (Grantham, 1986; Moldowan et al., 1991). The Yamama oils are grouped into two sub-families, with an average API gravity of 38°, suggesting a higher level of thermal maturity than the oils, which have an average API of 32°. Also, the Nahr Umr and Mishrif oil samples have a much lower average API value of 22°. These different physical properties may suggest that this oil probably originated from many different source rocks with different thermal maturities (Fig. 4).

Crude oils with high sulfur content (generally >1.0%) are typically associated with marine carbonates or evaporitic source rocks deposited under strongly reducing (anoxic) conditions (Koopmans et al., 1998). In contrast, low-sulfur oils often point toward siliciclastic source facies or environments with higher oxygen availability (Koopmans et al., 1998; Peters et al., 2005). All studied samples have Sulfur content higher than >1.0%, referring to marine carbonates deposited under anoxic conditions (Fig. 2). The trace elements vanadium (V) and Nickel (Ni) are incorporated into organic porphyrin complexes during early diagenesis and are remarkably stable throughout the migration and maturation history of the oil (Lewan, 1984). A high concentration of Vanadium relative to Nickel is a strong indicator of a marine carbonate source deposited in a highly anoxic environment (Barwise, 1990). The V/(V+Ni) ratio is a robust geochemical proxy for reconstructing paleo-redox conditions. Ratios exceeding 0.5 for all studied samples are diagnostic of strongly reducing marine environments (Peters et al., 2005).

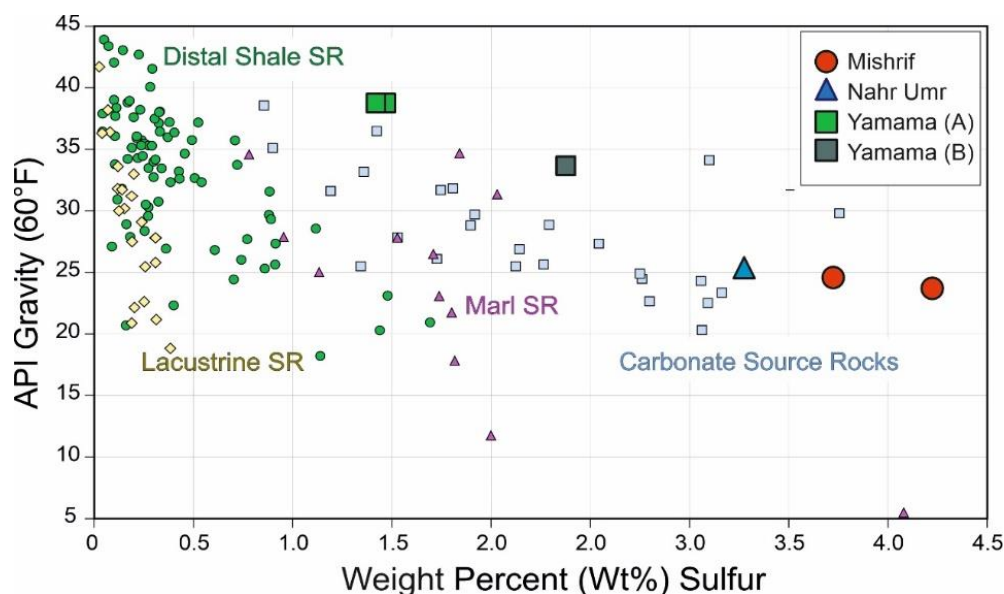


Fig.4. Average API Gravity and weight % sulfur of Yamama, Nahr Umr, and Mishrif reservoir crude oils from the Ratawi Oil Field, south of Iraq

The Yamama sub-family (A) oils have a lower V/Ni ratio (1.3) than the other studied oil samples, which average 3.6. Since both values significantly exceed the 0.5 threshold, they confirm a common origin from marine carbonate source rocks deposited under strongly reducing (anoxic) conditions. The 1.3 ratio represents a typical anoxic marine environment, possibly with slight water column fluctuations or a carbonate-marl lithology, consistent with the standard Lower Cretaceous or Middle Jurassic formations in the Mesopotamian Basin, south of Iraq. However, the significant difference between 1.3 and 3.6 suggested that the other samples were derived from different organic facies or stratigraphic potential source rocks (Magoon and Dow, 1994). As a result, while found in the same oil field, crude oils have a ratio of 3.6, which probably migrated from a deeper, more restricted part of the basin, possibly the Middle Jurassic Sargelu Formation. Where the 1.3 ratio suggests a more local charge from the Yamama or Sulaiy source rocks.

3.3. Stable Carbon Isotope Analysis ($\delta^{13}\text{C}\%$)

Stable Carbon Isotope analysis ($\delta^{13}\text{C}\%$) is an effective and widely used method for determining the genetic relationships between crude oils and their source rocks, as well as characterizing depositional environments and organic input (Sofer, 1984). A positive $\delta^{13}\text{C}$ value indicates that the sample is enriched (heavier) relative to the standard, while a negative value indicates that it is depleted in ^{13}C (lighter). (Peters et al., 2005).

Yamama oil samples are separated into two genetic sub-families depending on their stable carbon isotope ratios ($\delta^{13}\text{C}_{\text{sat}}$ and $\delta^{13}\text{C}_{\text{aro}}$) as shown in Fig. 3. Sub-family A has an enriched ratio value (average -26.45 for $\delta^{13}\text{C}_{\text{sat}}$ and -26.25 for $\delta^{13}\text{C}_{\text{aro}}$), suggesting a localized, self-sourced hydrocarbon charge from the Lower Cretaceous Yamama Formation. Yamama sub-family B has more depleted ratio values, averaging -28.1 for $\delta^{13}\text{C}_{\text{sat}}$ and -27.77 for $\delta^{13}\text{C}_{\text{aro}}$, indicating an important contribution from the Middle-Late Jurassic Sargelu Formation. These genetic classifications are consistent with the studies suggested by Sofer (1984) and Al-Ameri (2015). Oils accumulated in the Mishrif and Nahr Umr formations have intermediate isotopic signatures ($\delta^{13}\text{C} = -27.4$ and -27.35 , respectively), suggesting a mixed hydrocarbon source from Sargelu, Sulaiy, and Yamama or Sulaiy formations (Fig. 5).

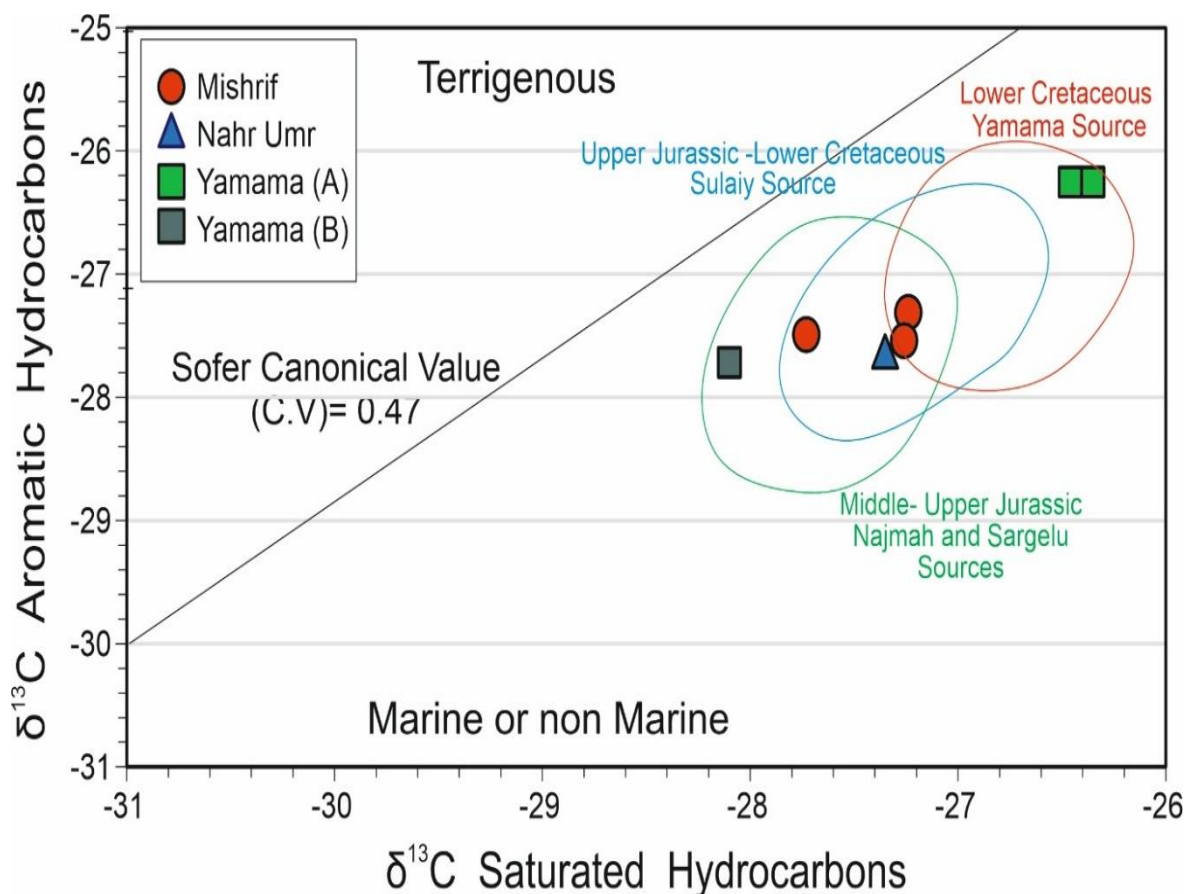


Fig. 5. Cross-plot of Stable Carbon Isotope Composition in mil PDB ($\delta^{13}\text{C}$ aromatic versus $\delta^{13}\text{C}$ saturated) for the Yamama, Nahr Umr, and Mishrif crude oil samples in the Ratawi Oil Field, south of Iraq

3.4. Biomarker Organic Geochemistry

Biomarkers, or biological markers, are complex organic molecules found in petroleum and source rocks that have retained the carbon skeleton of their biological precursors. These compounds are highly resistant to thermal degradation, allowing them to remain intact during diagenesis and catagenesis (Peters et al., 2005).

3.4.1. Alkanes and acyclic isoprenoid ratios

These ratios are a primary indicator of the redox conditions of the paleo-depositional environment of the source rock because it is influenced by source input and thermal maturity. Both pristane (C19) and phytane (C20) are acyclic isoprenoids derived from the degradation of phytol, a side chain of the chlorophyll molecule (Brooks et al., 1969). In oxygen-rich (aerobic) environments, phytol degrades to pristane. In oxygen-poor (anoxic) environments, it degrades to phytane. Thus, a low Pr/Ph ratio (< 1) indicates an anoxic marine or saline-lacustrine environment, while a high ratio (> 3) suggests an oxic, terrigenous setting (Bray and Evans, 1961). The high content of Pr and Ph in most crude oils enables direct analysis using gas chromatography traces rather than GCMS. Pr/Ph is widely used in correlation because it usually increases with thermal maturation (e.g., ten Haven et al., 1987). Fig. 4 shows that the Pr/n-C17 and Ph/n-C18 values (Fig. 6) indicate that the crude oil samples from the Yamama, Nahr Umr, and Mishrif reservoirs are non-biodegraded and mature oils, marine algal type II kerogen. The Yamama sub-family (A) oil samples are the most thermally mature.

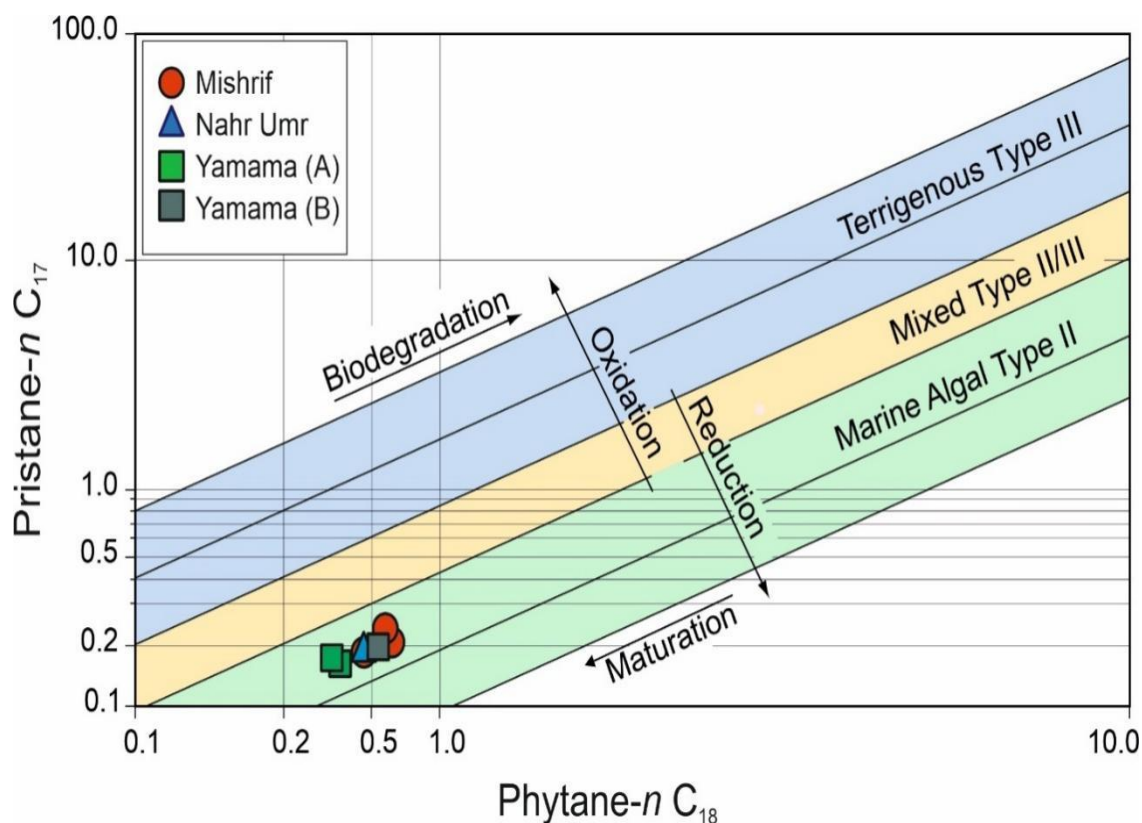


Fig. 6. Pristane/ n-C₁₇ Versus Phytane/ n-C₁₈ for crude oil samples (modified from Peters et al., 1999)

3.4.2. Carbon preference index (CPI)

These indices measure the dominance of odd-or even-numbered n-alkanes. This predominance originates from primary biological sources; terrigenous organic matter (e.g., plant waxes) often shows a strong odd-over-even predominance (e.g., C₂₉, C₃₁). With increasing thermal maturity, thermal cracking of these molecules produces a wider range of both odd- and even-numbered alkanes, causing the CPI and OEP values to trend towards 1.0. All studied crude oil samples have a CPI value of about 1.0, suggesting mature oils.

3.5. Terpanes and Hopanes

3.5.1. Tricyclic and tetracyclic terpanes (C₁₉ - C₃₀)

These low-molecular-weight compounds are mainly derived from the cell membranes of marine algae and bacteria (Peters et al., 2005). Tricyclic terpanes (TTs) are stable and resistant to degradation. Abundant TTs indicate preservation conditions in depositional environments rather than a unique biological source. The ratio of C₂₃ to C₂₄ tricyclic terpanes (TR₂₃/TR₂₄) is an indicator for identifying the depositional environment and organic matter source of crude oils (Zumberge, 1987).

All studied crude oil samples (Mishrif, Nahr Umr, and Yamama A & B) are located in the upper-left quadrant of the Tricyclic Terpane Plot (Fig. 7) and have C₂₂/C₂₁ ratios greater than 1.0, indicating carbonate source rocks deposited under reducing conditions. The Yamama samples have higher C₂₆/C₂₅ ratios (approaching 0.9), indicating a source facies with higher shale content than that of the Mishrif and Nahr Umr oils.

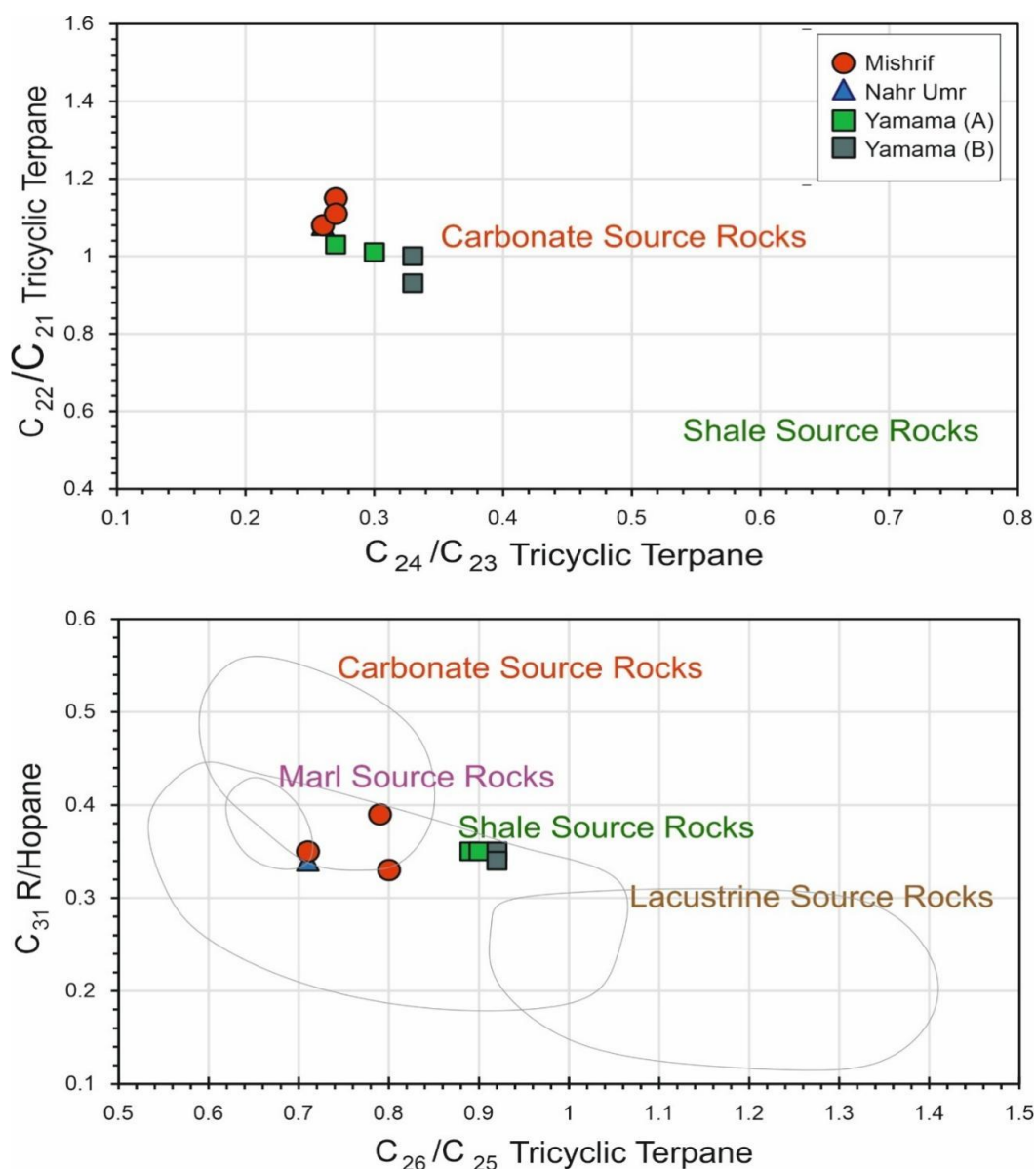


Fig. 7. Terpanes via the hopane relationship show the depositional environment of source rocks

All studied samples have low C_{19}/C_{23} ratios, which range from 0.12 in the Lower Cretaceous reservoirs (Nahr Umr/Mishrif) to 0.22 in the Yamama (A) sub-family. These low values always point to a minor contribution from terrestrial organic matter, indicating a marine microbial and algal source. C_{24}/C_{23} ratios are very low, ranging from 0.26 to 0.27; less than 0.5 indicates a highly diagnostic marine carbonate source rock (Tissot and Welte, 1984; Peters et al., 2005). This carbonate rock is further corroborated by the C_{22}/C_{21} ratios, which hover near unity (0.99 – 1.01), classic evidence of carbonate-evaporitic facies in the Mesopotamian Basin. The C_{26}/C_{25} ratios show moderate to high values, with the Yamama sub-family (A) yielding 0.89 compared to 0.75 for the other oil samples. The C_{26}/C_{25} ratios approaching 1.0 suggest that these carbonates were deposited in high-salinity marine environments. The slightly higher values in the Yamama (A) group suggest a more pronounced hypersaline influence during the deposition of its specific source facies. The differences in oil quality (API 38 vs 22) and redox (V/Ni 1.3 vs 3.6) are likely attributable to thermal maturity and anoxia, but the lithology remains a consistent high-salinity marine carbonate across all samples.

3.5.2. Pentacyclic terpanes (hopanes) C_{27} - C_{35})

Hopanes are the most diagnostic biomarkers for determining lithology and redox conditions. Derived from bacterial prokaryotes, they are highly stable during thermal maturation (Peters et al., 2005). The C_{29}/C_{30} hopane ratio is a critical Lithology Parameter; values exceeding 0.6 (Fig. 8) definitively point to marine carbonate source rocks. Furthermore, Homohopanes (C_{31} - C_{35}), specifically the C_{35} Index, are used to reconstruct the oxygen levels in the ancient bottom waters, where high values signify strongly reducing (anoxic) environments (Peters et al., 2005).

Hopane parameters, such as the C_{29}/C_{30} Hopane ratio (Fig. 6), differentiate between marine carbonate or evaporitic source rocks (ratios > 0.6) and siliciclastic origins (ratios < 0.5) (Peters et al., 2005). A C_{31}/C_{30} Hopane ratio above 0.25 confirms a marine source, while the $C_{35}S/C_{34}S$ Hopane Index indicates redox conditions, with values > 1.0 suggesting anoxic environments (e.g., Andrusevich et al., 2000). Gammacerane, a pentacyclic triterpane, identifies water column stratification from hypersalinity. Cross-plots reveal high C_{29}/C_{30} ratios for carbonates and low ratios for shales, while C_{35}/C_{34} and Gammacerane indices differentiate between anoxic/hypersaline and open shelf marine contexts.

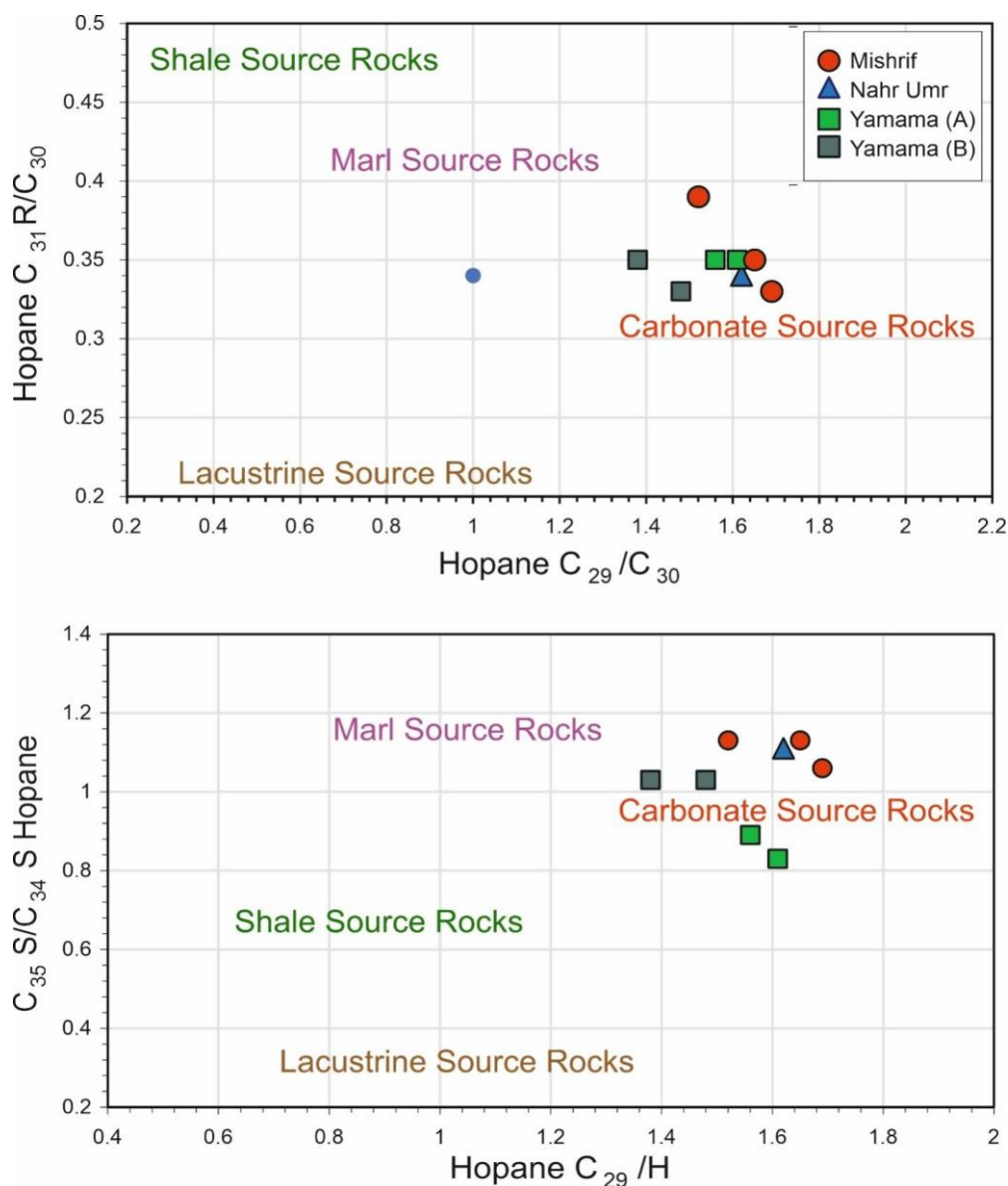


Fig. 8. Relationship diagrams between Terpanes parameters showing depositional environments and depositional environment (Modified from Zumberge, 1987)

In this study, the presence of Gammacerane, especially in the Nahr Umr and Mishrif samples with high V/Ni ratios (3.6), provides evidence that the potential source rock (likely the Sargelu) was deposited in a restricted, hypersaline marine environment, distinct from the more open-marine setting of the Yamama (A) oils. The high C₂₉/C₃₀ ratios on cross-plots (Fig. 8) indicate carbonates. The C₃₅/C₃₄ and Gammacerane indices assist in distinguishing between anoxic/hypersaline and open-shelf marine environments (Peters et al., 2005).

Fig. 9 illustrates that the C₃₁R/Hopane versus C₃₅/C₃₄S Hopane cross-plot shows a marine carbonate depositional environment for all studied samples, with a shift towards highly reducing conditions in the Mishrif, Nahr Umr, and Yamama (B) groups. This is also supported by the Gammacerane Index (GA/C₃₁R) vs. C₃₅/C₃₄ relationship (Fig. 7). The Mishrif and Yamama (B) oil samples show clustering in the hypersaline-anoxic depositional environment. However, the Yamama (A) sub-family has lower isomerisation and salinity indices, suggesting a distinct change in comparison to more open-marine, less restricted carbonate source facies (such as the Sulaib/Yamama sequence).

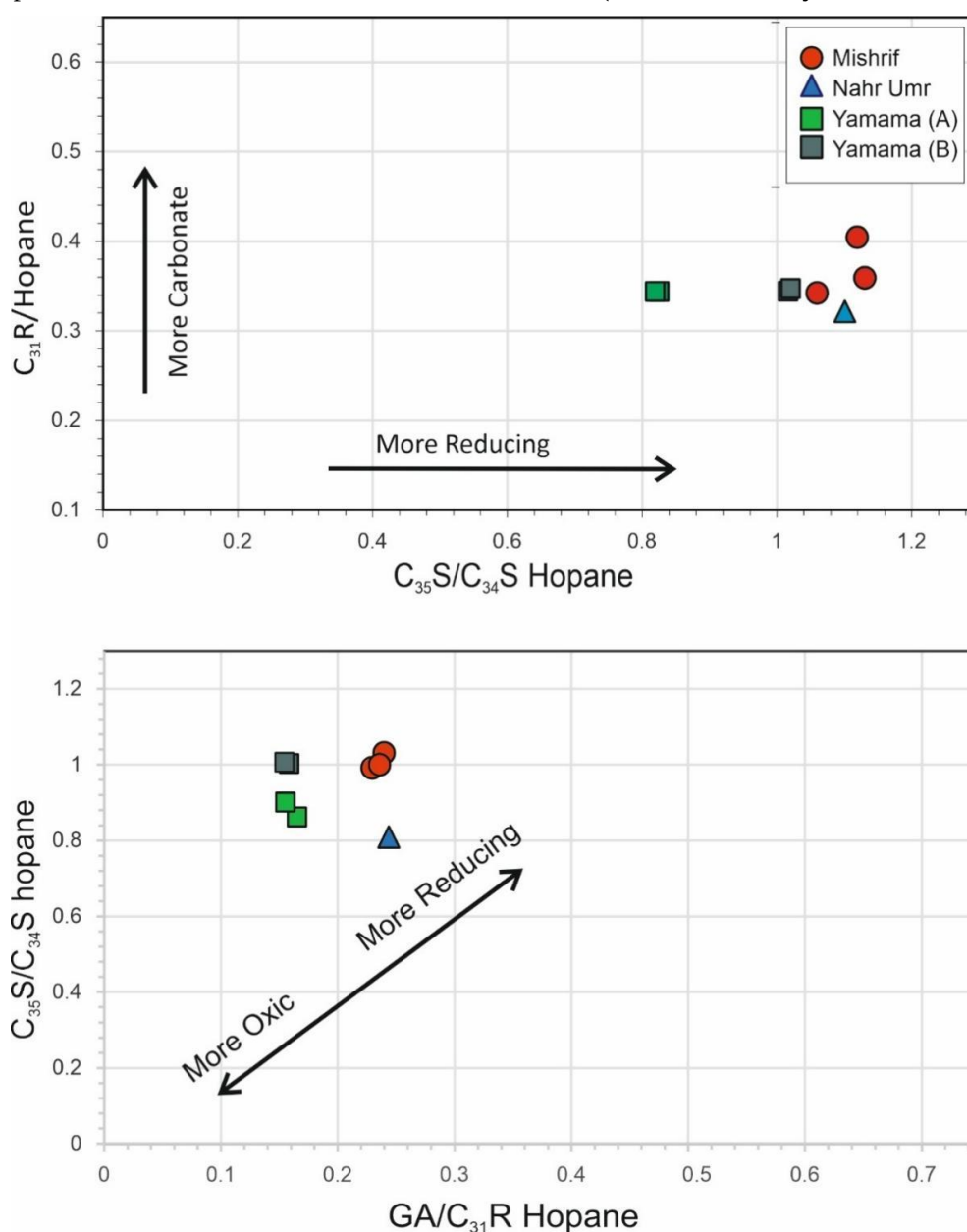


Fig. 9. Cross-plots of hopane biomarkers differentiate the studied oils into two distinct paleo-environmental regimes

The Oleanane in crude oil is a specific indicator of higher plant input (angiosperms). The oldest angiosperms are from the Early Cretaceous and younger (Peters et al. 2005). The presence of oleanane in trace amounts (Oleanane Index ranging from 0 to 0.01) in all studied samples confirms a Cretaceous organic contribution. This index implies a mixed-source origin for the high-maturity Yamama (A) sequence, with Middle Jurassic oils migrating upward through Cretaceous strata. By contrast, the Mishrif and Nahr Umr oils have molecular characterisation that clearly indicates a Lower Cretaceous genetic affinity.

3.6. Regular Steranes Distribution

The relative abundance of these steranes can be plotted on a ternary diagram to help differentiate between source rock environments and the type of organic input. For example, a high abundance of C_{27} steranes is always linked to an algal (marine) source, while C_{29} steranes are more associated with terrestrial plants (Peters et al., 2005).

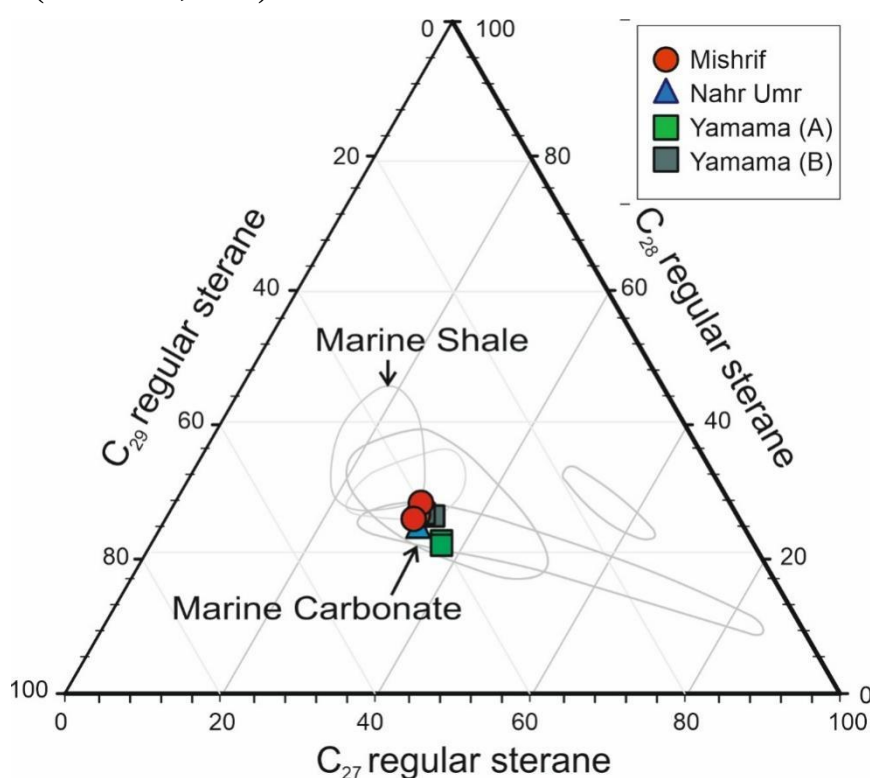


Fig. 10. A ternary diagram displays the relative abundances of C_{27} , C_{28} , and C_{29} regular steranes in saturated fractions of Ratawi crude oil (Modified from Peters et al., 2005)

Geochemical findings from the Ratawi Oil Field show two distinct oil families. This is most evident in the difference between the Yamama Sub-family (A) and the other reservoir fluids (Mishrif, Nahr Umr, and Yamama B). The Yamama (A) oils have a high API gravity of about 38° and a sterane distribution of C_{27} (37): C_{28} (22): C_{29} (41). The significantly lower C_{28} content (22%) may suggest a Jurassic Sargelu Formation source affinity. The Mishrif, Nahr Umr, and Yamama (B) samples have higher C_{28} percentages (~25%), which is typical of the Lower Cretaceous Sulaiy or Yamama formations' source rocks (Al-Ameri, 2015; Al-Khafaji et al., 2019) (Fig. 10). The C_{28}/C_{29} sterane ratios give important chronological evidence (Peters et al., 2005). The Yamama (A) has a lower ratio (0.55) that strongly aligns with a Jurassic source affinity, confirmed by its high thermal maturity (high T_s/T_m and Triaromatic Steroid (TAS) derived from a deeper source. In contrast, the higher ratios in the Mishrif, Nahr Umr, and Yamama (B) samples (avg. 0.61) indicate a younger, Lower Cretaceous source

age (such as the Sulaib/Yamama sequence), even with their highly reducing and hypersaline geochemical characterisation (Fig.11).

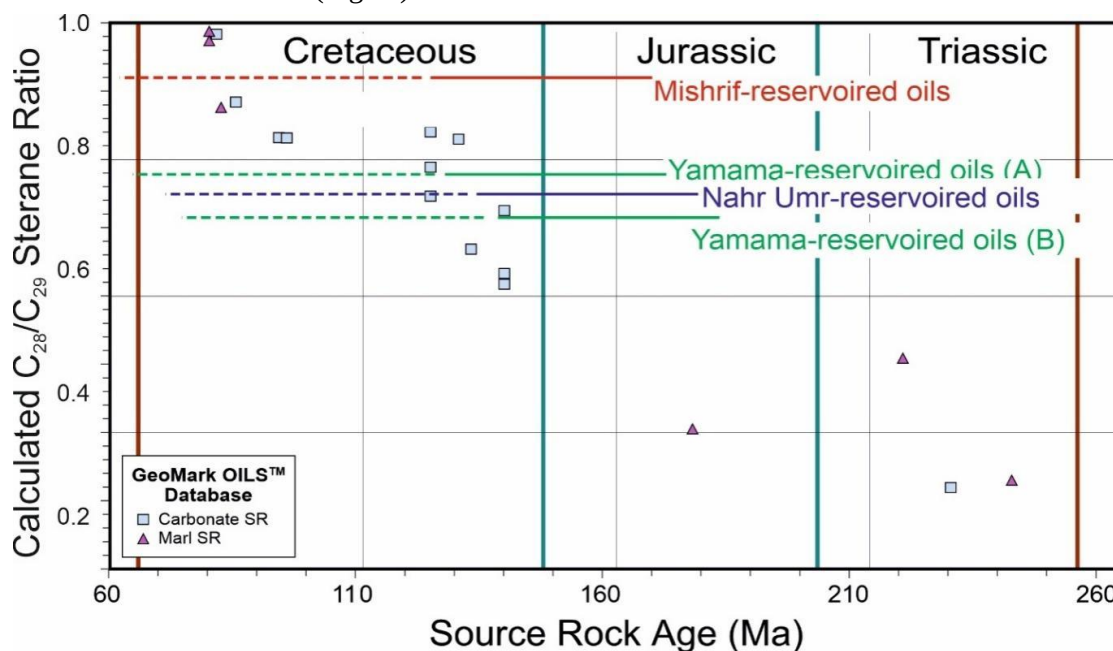


Fig. 11. Calculated C28/C29 sterane ratio of the studied oil samples from the Ratawi Oil Field

The Nahr Umr oils contain a higher concentration of C₂₉ steranes (~42.1%). The C₂₉ steranes can be generated by specific marine algae in carbonate depositional environments. Its high concentration in the Nahr Umr oil samples suggested a greater contribution from terrestrial higher plants. This suggests that the Nahr Umr source was situated closer to the shore, where terrestrial precipitation was more common.

3.7. Thermal Maturity Parameters

The ratio of Ts (trisanthrene) to Tm (trisanthrene) is a highly sensitive maturity indicator. As the thermal maturity level increases, the Tm becomes less stable and changes to Ts. This parameter is likely to confirm the 38° API results for the Yamama sub-family (A), which shows higher Ts/(Ts+Tm) values than the shallower, less mature Mishrif or Nahr Umr oil samples (22° API).

The thermal maturity of the Ratawi Oil Field was evaluated through sterane isomerisation and aromatic steroid distributions (Zumberge, 1987). The C₂₉ 20S/(20S+20R) ratios for all samples (0.60-0.63) show that the oils have reached or exceeded the thermal equilibrium point (approx. 0.55) (Peters et al., 1990). The Yamama (A) sub-family exhibits significantly higher values, C₂₇ Ts/Tm (0.74) and TAS (0.63), compared to the shallower Cretaceous reservoirs (C₂₇ Ts/Tm = 0.22; TAS = 0.30) (Fig. 12). These findings confirm that the Yamama (A) oils reached higher thermal maturity during the late oil window (TAS3 = 0.63; C₂₇ Ts/Tm = 0.74), while the Yamama (B), Nahr Umr, and Mishrif oils were generated in the early oil window (TAS3 approx. 0.3; C₂₇ Ts/Tm approx. 0.2). This maturity contrast explains the observed differences in API gravity (38° vs. 22°) and offers validity to the Yamama (A) subfamily's idea of a deeper, high-maturity source.

The Yamama (A) oils have higher δ¹³C values (-26.4‰) than the Jurassic sources in Iraq, which can be attributed to thermal cracking and the Kinetic Isotope Effect. According to Peters et al. (2005), increasing thermal maturity results in a progressive enrichment of δ¹³C in the residual oil fraction. This isotopic change, along with the high TAS and T/Tm ratios, confirms that the Yamama (A) has a highly developed hydrocarbon charge.

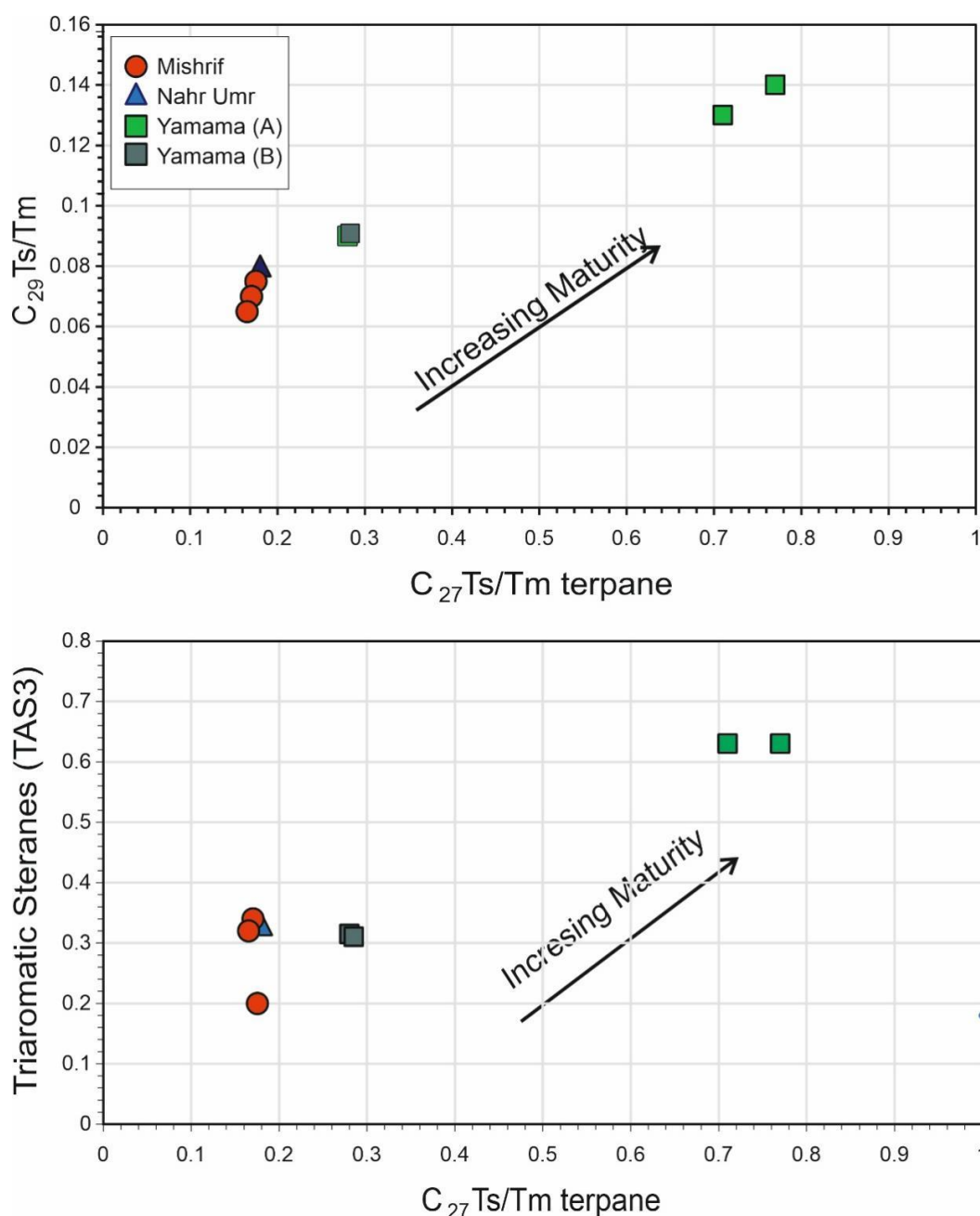


Fig. 12. Thermal Maturity Assessment based on Terpane and Sterane Rearrangements of the studied oil samples

4. Conclusions

The tectonic timing between the peak generation of the Jurassic-Cretaceous source rocks and salt-driven structural growth resulted in a highly effective charging mechanism. Geochemical and tectonic integration define two primary genetic oil families in the Ratawi Oil Field: (1) Family I (Middle Jurassic-Distal Marine) is mainly found in the Yamama (A) reservoir and is distinguished by its organisation from the Middle Jurassic Sargelu/Naokelekan generating pods. Higher thermal maturity due to high TAS, $C_{27}ts/Tm$ ratio, and CPI near 1.0, low Hopane/Sterane ratio (<1.0), low Pr/Ph (<1.0), and C_{27} dominance indicate a distal, anoxic carbonate environment. The lower C_{28}/C_{29} sterane ratio (0.55) suggests the Middle Jurassic Sargelu Formation. Vertical migration may be charged from deep basinal depocenters via fault systems caused by the Hormuz Salt Halokinesis, and (2) Family II (Lower Cretaceous - Proximal/Transitional): Identified in the Mishrif, Nahr Umr, and Yamama (B) reservoirs,

this family exhibits unique geochemical characteristics linked to the Lower Cretaceous source facies with mixed marine and terrigenous input.

Family II (Lower Cretaceous - Proximal/Transitional): Identified in the Mishrif, Nahr Umr, and Yamama (B) reservoirs, it displayed geochemical characteristics related to the Cretaceous source facies, which received mixed marine and terrigenous inputs. Higher C_{28}/C_{29} sterane ratios (> 0.6) indicate the existence of oleanane, and high abundances of C_{24} tetracyclic terpanes, together with high hopane/sterane ratios and a distinct gammacerane index, suggest a nearby marine depositional environment, characterized by significant bacterial reworking and water-column stratification. Also, a significant enrichment of C_{29} steranes (up to 42%) in the Nahr Umr reservoir displays a strong proximity to land-derived organic matter.

Author contributions: CRediT

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Declaration of Competing Interests

The authors declare that they have no competing financial interests or personal relationships that could influence the work reported in this study.

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Data Availability

The data used in this study were obtained from a third party and are not publicly available. Access to these data may be obtained from the data owner, according to their restrictions.

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